

HyperCom Overview

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```
library(HyperCom)
library(dplyr)
library(ggplot2)
library(ggribes)
library(Matrix)
library(tidyr)
```

Load in example data

HyperCom requires three inputs:

1. A gene x cell matrix with genes as row names and cells as column names
2. A dataframe of cell metadata containing sample and optionally group columns which need to be specified as variables
3. A dataframe of ligand-receptor (LR) interactions. HyperCom contains 2 built in LR databases formatted properly:
 - CellChat human `cellchat.hs.lrd`
 - CellChat mouse `cellchat.mm.lrd`

In this vignette we will use a 1000 cell subset of heart failure from [Alexanian et al, 2021](#). This example subset is included in the package as `alexanian2021.counts` and `alexanian2021.metadata`.

```
counts <- alexanian2021.counts
dim(counts)
```

```
[1] 17083 1000
```

```
head(rownames(counts))
```

```
[1] "Xkr4" "Sox17" "Mrpl15" "Lypla1" "Tcea1" "Rgs20"
```

```
head(colnames(counts))
```

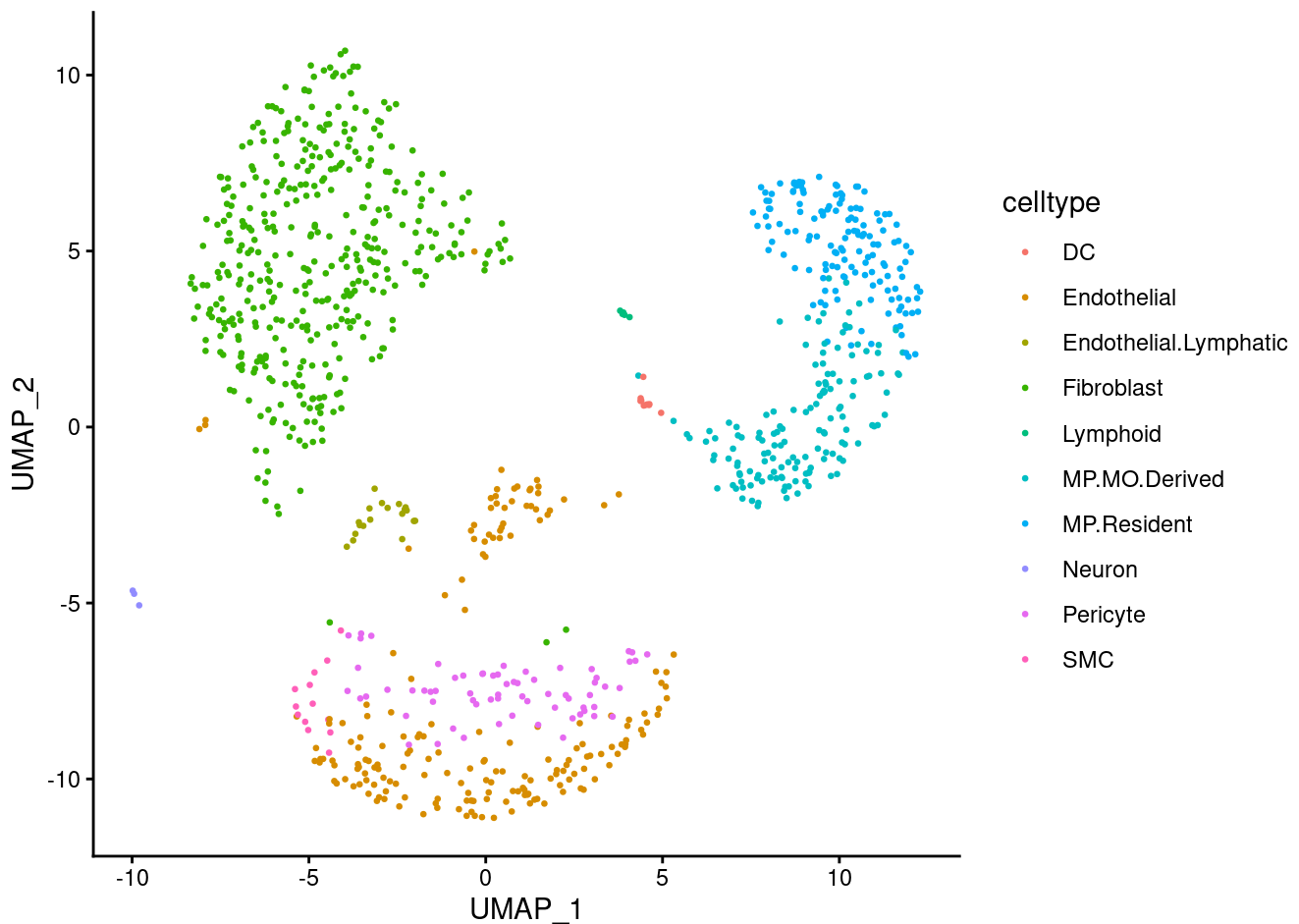
```
[1] "AACCGCGTCCAGAGGA-2" "AACTCAGCACGAGGTA-2" "AACTCTTGTGCGAAAC-2"
```

```
[4] "AAGGAGCAGGATCGCA-2" "ACACCAAAGACCCACC-2" "ACACCGGAGCTACCGC-2"
```

```
meta.data <- alexanian2021.metadata
sample.column <- "sample"
group.column <- "condition"
head(meta.data)
```

	sample	condition	celltype	UMAP_1	UMAP_2
AACCGCGTCCAGAGGA-2	2	SHAM	Endothelial	1.2676813	-10.036637
AACTCAGCACGAGGTA-2	2	SHAM	Endothelial	1.7707077	-9.448688
AACTCTTGTCGAAAC-2	2	SHAM	Endothelial	0.2925013	-9.781118
AAGGAGCAGGATCGCA-2	2	SHAM	Endothelial	1.1058826	-10.454998
ACACCAAAGACCCACC-2	2	SHAM	Endothelial	-3.3748763	-10.034042
ACACCGGAGCTACCGC-2	2	SHAM	Endothelial	3.9700553	-8.899605

```
ggplot(meta.data, aes(x=UMAP_1, y=UMAP_2, colour=celltype)) +
  geom_point(size=0.5) +
  theme_classic()
```



For the LR interactions we will use the CellChat murine database.

```
lrdb <- cellchat.mm.lrdb
dim(lrdb)
```

```
[1] 2377    6
```

```
head(lrdb, 2)
```

```
      interaction ligand      r1      r2  r3  r4
1 TGFB1_TGFB1_TGFB2  Tgfb1 Tgfbr1 Tgfbr2 <NA> <NA>
2 TGFB2_TGFB1_TGFB2  Tgfb2 Tgfbr1 Tgfbr2 <NA> <NA>
```

Running HyperCom

The standard pipeline can be run using the [HyperCom](#) function. This vignette details each component function of the pipeline.

```
HyperCom(lrdb=cellchat.mm.lrdb, counts=alexanian2021.counts,
          metadata=alexanian2021.metadata, sample="sample",
          group="condition", keep=NULL,
          significance=FALSE, parallel=FALSE, cl=NULL)
```

Preprocessing

In order to reduce memory usage, we first filter the counts matrix to only genes present in the LR interactions with expression greater than 0. The function also transposes the matrix. We can also use the [load_seurat](#) function to directly complete this step from a Seurat v5 object.

```
lr.counts <- filter_counts(counts=counts, lrdb=lrdb, outfile=NULL)
dim(lr.counts)
```

```
[1] 1000  663
```

Next we filter the ligand receptor interactions to only interactions where all genes are present in the filtered expression matrix.

```
lrs <- generate_lrs_table(counts=lr.counts, lrdb=lrdb, outfile=NULL)
dim(lrs)
```

```
[1] 1042    5
```

Generating Hypergraph

To generate the hyperedges, which are edges that contain multiple nodes, we use the expression matrix LR interaction to create 2 types of hyperedges:

1. expression edges - edges containing cells that express a given ligand or receptor
2. interaction edges - edges containing the genes that comprise an LR interaction

HyperCom represents these hyperedges as a dataframe with an edge and node column where the node belongs to the edge. The weight is set to 1 for each edge.

```
hyperedges <- generate_hyperedges(counts=lr.counts, lrs=lrs, outfile=NULL)
head(hyperedges, 2)
```

```
# A tibble: 2 × 3
```

	edge	node	weight
	<chr>	<chr>	<dbl>
1	TGFB1_TGFBR1_TGFBR2	Tgfb1	1
2	TGFB1_TGFBR1_TGFBR2	Tgfbr1	1

We can represent this hypergraph as an adjacency matrix where the nodes are the cells, ligand genes, and receptor genes. Each entry is the number hyperedges shared by nodes.

```
adj <- generate_adjacency_matrix(hyperedges=hyperedges, outfile=NULL)
dim(adj)
```

```
[1] 1663 1663
```

Calculating Network Connectivity

In order to calculate the network connectivity of the hypergraph we perform a random walk on the adjacency matrix to produce an influence matrix.

```
infMat <- generate_infMat(adj=adj, outfile=NULL)
dim(infMat)
```

```
[1] 1663 1663
```

Downstream Analysis

To account for genes now being present in the matrices, we then process the metadata to contain information on the genes as well as cell metadata. Additionally to reduce memory, we reduce the number of columns to just node, sample, and group. Optionally other metadata columns can be kept.

```
metadata <- generate_metadata(infMat=infMat, metadata=meta.data,
                             sample=sample.column, group=group.column,
                             keep=c(), outfile=NULL)

head(metadata)
```

	cell	sample	group
1	AAAGATGCACAGATTC-3	3	TAC
2	AAAGATGCAGCTCGCA-3	3	TAC
3	AAAGATGCAGGTCTCG-3	3	TAC
4	AAAGATGGTGTGTTGGT-2	2	SHAM
5	AAAGTAGAGGAGTTTA-2	2	SHAM
6	AAAGTAGCACATTTCT-2	2	SHAM

Identifying Important LR Pairs

We can also identify important LR pairs by using HyperCom scoring on LR pairs. This prioritization scores LR pairs to find highly communicative LR pairs that are optionally group specific. A p-value can be calculated for each LR pair, but doing so will vastly increase the time to run. By default the `significance` arg is FALSE. The `prioritize_lr` function can be run sequentially or parallelly and by default the `parallel` arg is FALSE. The parallel implementation uses the `parallelly` package, but the `cl` can be supplied any object of class `cluster` depending on your system's architecture.

```
priority <- prioritize_lr(lrs=lrs, adj=adj, infMat=infMat, metadata=metadata,
                         significance=FALSE, parallel=TRUE, cl=NULL, outdir=NULL)

head(priority)
```

	interaction	ligand	r1	r2	r3	priority
1	THBS1_SDC4	Thbs1	Sdc4	<NA>	<NA>	0.3313764
2	PGF_VEGFR1	Pgf	Flt1	<NA>	<NA>	0.3179562
3	COMP_SDC4	Comp	Sdc4	<NA>	<NA>	0.2923955
4	FN1_SDC4	Fn1	Sdc4	<NA>	<NA>	0.2717061
5	POSTN_ITGAV_ITGB5	Postn	Itgav	Itgb5	<NA>	0.2651227
6	CCL4_CCR5	Ccl14	Ccr5	<NA>	<NA>	0.2645498

We see that Thbs1-Sdc4 is the highest scoring LR pair. We can calculate HyperCom scores for this pair. As a note, the `score_hypercom` function can also take a vector of receptor genes if the receptor is a complex.

```
Thbs1_Sdc4 <- score_hypercom(infMat=infMat, metadata=metadata, ligand="Thbs1",
                             receptors="Sdc4")

head(Thbs1_Sdc4[, c(1, 2, 3, 6, 7)])
```

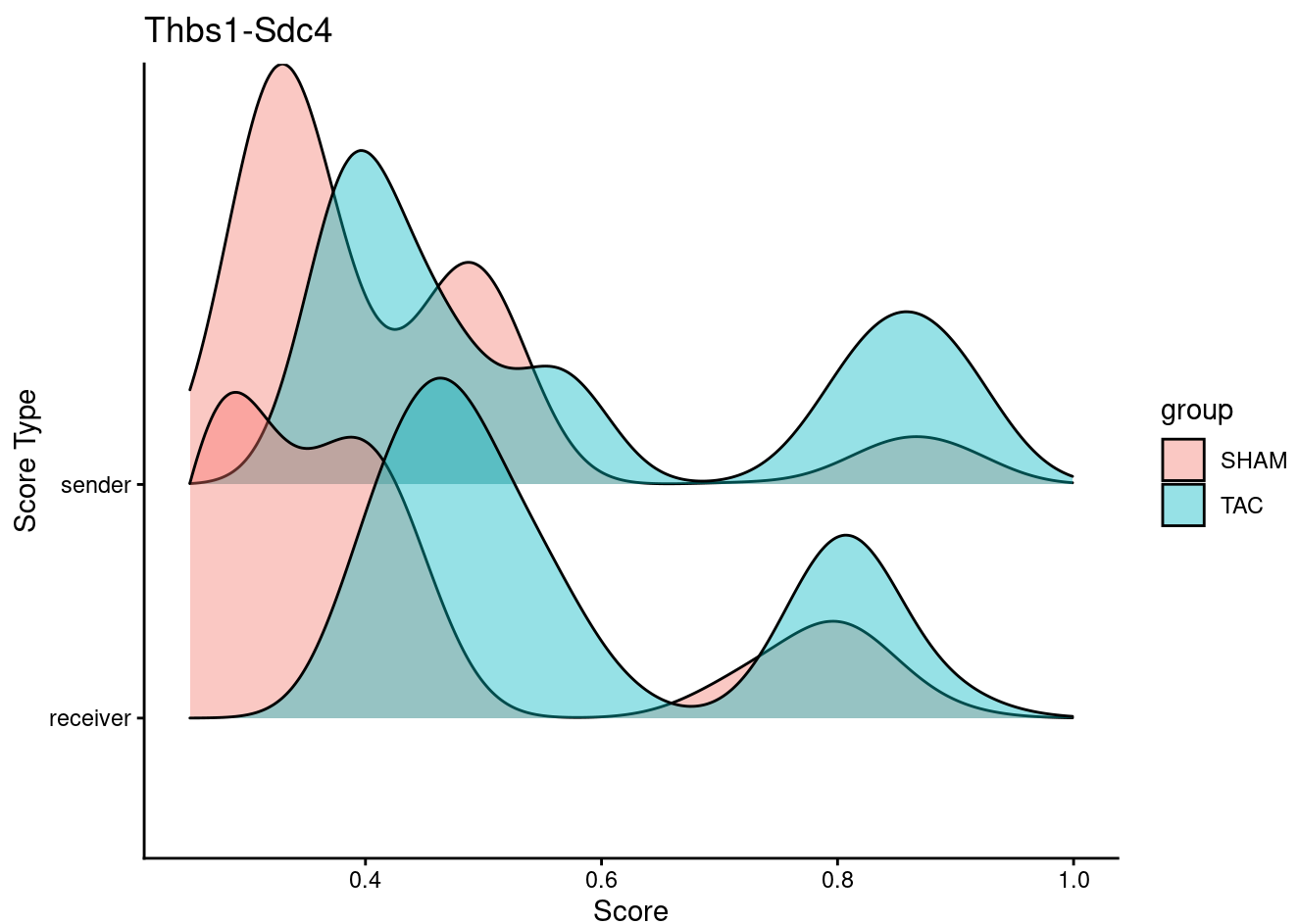
	cell	sender	receiver	max	hypercom
1	ATTATCCAGCTACCGC-3	0.4554808	0.9505336	0.9505336	0.6882292
2	CGTTAGAGTTACGCGC-3	0.4754217	0.9072661	0.9072661	0.6345970
3	ACTGTCCCAATAACGA-2	0.4947958	0.9135973	0.9135973	0.6245427
4	TCTCTAATCTTTAGTC-2	0.3524036	0.8681149	0.8681149	0.6053723
5	CTGAAGTAGTCGTTTG-3	0.3960002	0.8620137	0.8620137	0.6044511
6	CATTATCTCTCTGCTG-2	0.3319691	0.8752518	0.8752518	0.6017978

If we plot the sender and receiver score we see both higher densities of sender and receiver cells in TAC.

```
Thbs1_Sdc4 <- left_join(Thbs1_Sdc4, metadata, by=join_by(cell))
melt.Thbs1_Sdc4 <- pivot_longer(dplyr::select(Thbs1_Sdc4, c(cell, group, sender, receiver)), !c(cell, group), values_from=score)

ggplot(melt.Thbs1_Sdc4, aes(x = score, y = type, fill=group)) +
  geom_density_ridges(alpha=0.4) +
  xlim(0.25, 1) +
  xlab("Score") +
  ylab("Score Type") +
  ggtitle("Thbs1-Sdc4") +
  theme_classic()
```

Picking joint bandwidth of 0.0387



```
sessionInfo()
```

R version 4.4.0 (2024-04-24)

Platform: x86_64-pc-linux-gnu

Running under: AlmaLinux 8.10 (Cerulean Leopard)

Matrix products: default

BLAS/LAPACK: FlexiBLAS NETLIB; LAPACK version 3.12.0

locale:

```
[1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
[3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
[5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
[7] LC_PAPER=en_US.UTF-8     LC_NAME=C
[9] LC_ADDRESS=C             LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
```

time zone: America/New_York

tzcode source: system (glibc)

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

other attached packages:

```
[1] tidyr_1.3.2      Matrix_1.7-0      ggribges_0.5.7
[4] ggplot2_4.0.0    dplyr_1.1.4       HyperCom_0.0.0.9000
```

loaded via a namespace (and not attached):

```
[1] gtable_0.3.6      jsonlite_2.0.0      compiler_4.4.0
[4] Rcpp_1.1.0        tidyselect_1.2.1    collapse_2.1.7
[7] parallel_4.4.0    dichromat_2.0-0.1   scales_1.4.0
[10] yaml_2.3.10       fastmap_1.2.0       lattice_0.22-6
[13] R6_2.6.1          labeling_0.4.3      generics_0.1.4
[16] knitr_1.50        iterators_1.0.14     htmlwidgets_1.6.4
[19] tibble_3.3.1      zigg_0.0.2          pillar_1.11.1
[22] RColorBrewer_1.1-3 rlang_1.1.6         utf8_1.2.6
[25] xfun_0.52         S7_0.2.0            RcppParallel_5.1.11-2
[28] doParallel_1.0.17 cli_3.6.5           withr_3.0.2
[31] magrittr_2.0.5    foreach_1.5.2       digest_0.6.37
[34] grid_4.4.0        rstudioapi_0.17.1   lifecycle_1.0.5
[37] vctrs_0.6.5       Rfast_2.1.5.2       evaluate_1.0.5
[40] glue_1.8.1        farver_2.1.2        codetools_0.2-20
[43] parallelly_1.46.1 rmarkdown_2.31      purrr_1.2.0
[46] tools_4.4.0       pkgconfig_2.0.3     htmltools_0.5.8.1
```